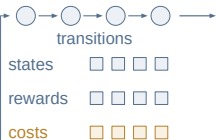


Observe

Policy rollout



Episode costs

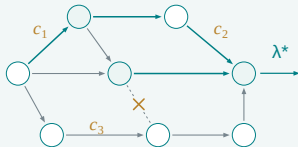


GAE advantages

Schematic

Construct shared prices

Joint graph



Solve jointly

Balance · per-step budgets
Newton / bisection

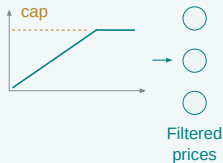
Episode-cost feedback



cost - budget

Accumulate

Clip + smooth



× hard graph support

β

Filtered prices

Update policy

Reward advantage

A_r

Cost advantages

A_c

Shared prices

×

+

-

Normalize

PPO / TRPO

π

next rollout